Bridging the gap risk reloaded: a comprehensive methodology for wrong way risk and leveraged exposures *

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Abstract

In the following, Fabrizio Anfuso extends the model for WWR and leveraged exposures introduced in [1] to non-linear payoffs and mixed portfolio compositions. For the generalised model, the author develops an analytical framework inclusive of: i) a method to embed WWR dynamics within existing Monte Carlo (MC) Counterparty Credit Risk (CCR) implementations; and ii) a closed-form expression for the Exposure at Default (EAD) in the presence of leverage and WWR that can be used for non-simulation based CCR portfolio valuations, such as stress testing.

1 Introduction

The sheer losses accrued by a handful of banks during the Archegos default [2] casts doubt on the applicability of the Basel Margin Period Of Risk (MPOR) framework to highly levered collateralised counterparties (CPTYs). Primarily, this is because such framework is unable to account for the structural WWR introduced by the contractual process of margining.

As discussed in a variety of contributions (see e.g. [1], [3], [4], [5], [6], [7]), the loss distribution in the presence of WWR and leverage is not exclusively characterised by statistical fluctuations of order $\sigma\sqrt{\text{MPOR}}$ on top of a smooth default; and any attempt to quantify realistically the closeout exposures depends crucially on the modelling assumptions regarding tail risk correlations; which in turn may not be directly observable, since they are driven by the overall risk position of the CPTY.

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Measuring risk with incomplete information is indeed one of the central conundrums when modelling the exposures of Non Bank Financial Institutions (NBFI) such as hedge funds. And international regulators post-Archegos ([8], [9], [10]) set an expectation for addressing this challenge either by achieving a better understanding of the true risk profile of the CPTY, or by requiring '*sufficiently conservative terms for the relationship if the client does not meet appropriate levels of transparency*" [9]. From a modelling perspective, this means primarily a stronger focus on stress testing metrics, indicated by a growing industry consensus that these are the only ones suitable for the risk monitoring of highly levered NBFIs and, especially, hedge funds [11].

In this spirit, **the main contribution of the present paper is an analytical framework that can be used to assess the impact of WWR and leverage on CCR exposures**; supporting the expert judgement of credit officers with quantitative metrics more reflective of the perceived risk profile of the CPTY.

The paper is structured as follows: in Secs.2 and 3 we define the reference model and metrics; in Sec.4, we benchmark the model predictions vs. Credit Suisse (CS) losses post Archegos default; in Sec.5, we introduce a perturbative expansion method to compute the collateral shortfall at default; in Secs. 6 and 7, we use the perturbative expansion to obtain a closed-form solution for the EAD and we apply our result to a number of relevant examples, including linear and non-linear payoffs; finally, we draw conclusions in Sec. 8.

2 The model

Consider two CPTYs, A and B, sharing a portfolio of derivatives contracts under the same netting agreement with daily bilateral variation margin (VM). Our aim is to compute the loss distribution of the surviving CPTY A conditional upon B's default; and we introduce a model for the correlated dynamics of the A-B portfolio and of the B's balance sheet Θ according to the following:

1. The portfolio MtM(t) is made of two components: i) $g(S(t), \sigma_S(t))$, denoting the fair value of the derivatives with underlying S (implied volatility σ_S), subject to gap risk and whose dynamics is correlated with Θ; and $V(t)$, denoting the fair value of the derivatives whose underlyings and prices are neither correlated with Θ or with S.

2. The *legacy* balance sheet Θ (before entering the A-B derivatives portfolio) is defined as for [1]. In our stylised model: i) Θ is stochastic with constant liabilities L, i.e. B cannot raise new debt to meet margin calls (this is realistic to model portfolio gap moves that may directly affect

the solvency of B); ii) the PD at the forecasting horizon T is considered as an input; iii) for the computations considered throughout the paper, $T = 1$y.

3. *Before entering the A-B derivatives portfolio*, the PD, L and $\Theta(0)$ are related[1] since L marks the threshold for the *legacy* Θ that triggers the B's default. Therefore, we consider $\Theta(0)$ as given and we compute the liabilities numerically; implicitly defining the starting Net Asset Value (NAV) as $\mathrm{NAV}(0) = \Theta(0) - L$.

4. *After entering the A-B derivatives portfolio*, the *new* Θ includes VM, that tracks the performance of the portfolio and can increase or reduce the NAV. Specifically, when the performance is in favour of A, since no new debt can be raised, any incremental collateral requirement should be met using the NAV resources; and hence the default time of B is now identified as the first $t > 0$ such that $\mathrm{NAV}(t) < 0$, i.e. B is unable to meet its pre-existing liabilities or a new margin call.

Based on the above, the joint dynamics of S, σ_S and Θ is described by the following system of SDEs:

$$
\begin{aligned}
\mathrm{MtM}(t) &= g(S(t), \sigma_S(t)) + V(t) && (1)\\
dS(t) &= \mu_S S(t)dt + \beta_{\sigma_S}\sigma_S(t)S(t)dW_S + S(t)dJ_S(t) && (2)\\
d\Theta(t) &= \mu_\Theta \Theta(t)dt + \sigma_\Theta \Theta(t)dW_\Theta(t) + \rho_\theta \Theta(t)dJ_S(t) && (3)\\
d\sigma_S(t) &= \frac{\rho_{\sigma_S}}{\mu_{J_S}}\bar{\sigma}_S dJ_S(t), && (4)\\
\Delta_{1d}(C_{\mathrm{VM}}(t)) &= -\Delta_{1d}(\mathrm{MtM}(t)),\ C_{\mathrm{VM}}(0) = -\mathrm{MtM}(0) && (5)\\
\Delta_{1d}(\mathrm{NAV}(t)) &= \Delta_{1d}(\Theta(t)) + \Delta_{1d}(\mathrm{MtM}(t)),\\
&\quad \mathrm{NAV}(0) = \Theta(0) - L + \mathrm{MtM}(0), && (6)
\end{aligned}
$$

where: i) The S and Θ dynamics are described by the Merton Jump-Diffusion model, both S and Θ are martingales and their Brownian drivers have instantaneous correlation ρ_W; ii) J_S is a compound Poisson process with constant jump size providing the correlated "gap risk" components of S, σ_S and Θ; iii) the implied volatility σ_S is purely driven by jumps and, for the purpose of this analysis, at most one jump can occur over the considered forecasting horizons; iv) $C_{\mathrm{VM}}(t)$ indicates the VM collateral balance from A's perspective, i.e. $C_{\mathrm{VM}}(t) < 0$ means that the trade is in the money for B.

For the A-B portfolio, we can express the leverage K_B and the initial margin (IM) of the B position (IM is not included in our model but we use

[1]Referring to the definitions provided in Sec.6, App.B and on the back of Eq.3, $\mathcal{M}(x = L, x_0 = \Theta(0), \mu_x = 0, \sigma_x = \sigma_\Theta, T = 1y) = \mathrm{PD}$, where this relation accounts only for the continuous dynamics of Θ.

it as benchmark) as:

$$
\begin{aligned}
K_B &= \frac{\sqrt{(g_{\mathrm{JTD}} - g(S(0), \sigma_S(0)))^2 + (q(1\%, \Delta V(0, \mathrm{MPOR})))^2}}{\Theta(0) - L} && (7)\\
\mathrm{IM} &= \sqrt{(q(1\%, \Delta g(0, \mathrm{MPOR})))^2 + (q(1\%, \Delta V(0, \mathrm{MPOR})))^2} && (8)\\
\omega_c &= \frac{g_{\mathrm{JTD}}^2}{g_{\mathrm{JTD}}^2 + (q(1\%, \Delta V(0, \mathrm{MPOR})))^2} && (9)
\end{aligned}
$$

where: i) $K_B(\Theta(0) - L)$ is a conservative measure of risk generally exceeding the IM, unless specific WWR and/or concentration add-ons apply; ii) g_{JTD} indicates the $t = 0$ Jump-To-Default (JTD) value of g; and iii) $q(1\%, \Delta V(0, \mathrm{MPOR}))$ indicates the 1st percentile of the distribution $\Delta V = (V(\mathrm{MPOR}) - V(0))$, aligned with the SIMM definition for diversified portfolios; iv) ω_c is a concentration measure that we use, in conjunction with K_B, to fully characterise the portfolio risk profile (deriving g_{JTD}, as a function of the notional of the trade, and $q(1\%, \Delta V(0, \mathrm{MPOR}))$, as a function of the characteristics of V, using Eqns.7 and 9).

3 The collateral shortfall $\mathrm{SF}_{\mathrm{VM}}$ and continuous time margining

As discussed extensively in [1], the standard Basel MPOR approach cannot account for margin driven defaults; where the structural WWR introduced by the contractually agreed process of margining may cause significant collateral shortfalls at default and materially affect the EAD.

Following [1] and focussing on VM-only contractual relations, we define the VM shortfall as:

$$
\mathrm{SF}_{\mathrm{VM}} = \mathbb{E}\big[\max\{-\Delta_{1\mathrm{d}}(\mathrm{MtM}(t^*)), 0\} | \mathcal{D} \neq \emptyset\big], \ \mathcal{D} = \{t_n \leq T | \mathrm{NAV}(t_n) < 0\}, \tag{10}
$$

where $t^* = \inf(\mathcal{D})$ indicates the first (if any) time on a path when, accounting for the margining obligations, the $\mathrm{NAV}(t)$ becomes negative.

Notice that Eqns.1-6 are defined for daily re-margining frequency. Going forward, we turn to a continuous time model, where the collateral is rebalanced instantaneously. While the two approaches are not equivalent from a risk stand point, they show the same qualitative dependencies on the key risk factors and are close enough for an indicative quantitative comparison.

4 The Archegos default: model forecast vs. realised loss

General considerations on how to calibrate *ab initio* Eqns.1-6 are provided in [1]. In this section, we test the plausibility of the forecasted collateral shortfalls vs. the closeout losses observed during the Archegos default, and specifically for the case of CS. Referring to the public information available in [2], we calibrate the model as follows:

1. The PD is set to $\approx$ 300bp, consistent with Archegos B+ rating as of 2021.

2. Archegos - CS portfolio is approximated with a single long stock position of $\approx$ \$7bn notional in the WWR underlying S. The jump component of S is calibrated based on the VIACOM stock that triggered the first margin calls and the disordered liquidation of the Archegos positions across the street; specifically: $\mu_{J_S} = -0.5$, mirroring VIACOM down move in March 2021, and $\lambda_{J_S} = 0.03$, compatible with the forecasted PDs for the media sector in 2021.

3. The leverage K_B is not known accurately but most likely in the range 0.5-1; hence we show results for a wide range of values.

4. The WWR component ρ_Θ is a free parameter of the model; based on *ex post* information, ρ_Θ is in the range 0.7-1 (*ex ante*, the correlations between NAV and portfolio returns, largely positive for Archegos, could have been used to estimate ρ_Θ).

5. The diffusion volatilities of the single stock portfolio and of the Archegos balance sheet (σ_S and σ_Θ) are set to 50% (annualised), slightly below the reported 70%, since a large contribution to the latter figure comes from large 'jump" like returns[2]. The diffusion correlation between the Brownian drivers of S and Θ is set to $\rho_W = 0.5$.

Fig.1 shows the model Potential Future Exposure (PFE) over 1y horizon computed at the 95th and the 99th quantiles. As discussed more extensively in Sec.6 in the context of the EAD, we approximate the PFE with its main contributor, i.e. the detected collateral shortfall at default (computing a quantile, rather than the positive average used for $\mathrm{SF}_{\mathrm{VM}}$ in Eq.10).

At the considered levels of confidence: i) the model PFEs are in the range \$2.5bn - \$4.1bn, to be compared with CS realised closeout loss of \$5.5bn; ii) the model PFEs are generally much larger than the IM for all the three IM benchmarks considered; and iii) the collateral shortfall at default (and hence the PFE) is driven by the interplay between leverage (K_B) and WWR

[2]Here the limitation of the model mentioned in Sec.2, i.e. the non-stochasticity of the jumps, comes into play; the VIACOM stock over 2020 - 2021 experienced a significant number of large moves, that cannot be explained by a purely diffusive behaviour but are still much smaller than the single jump size $\mu_{J_S} = -0.5$ set in the calibration. Since we have no lean way to calibrate the model for those, we ignore their contribution and ultimately reduce the volatility. The alternative, i.e. to increase the diffusion volatility, would tend to deplete $\mathrm{SF}_{\mathrm{VM}}$ because of an excessive contribution from $J_{1/W}$ paths (as defined in Sec.5).

(ρ_Θ), where extreme levels of either one or the other can also independently cause significant losses.

Notice finally that, as a function of ρ_Θ and of K_B, the probability of default increases significantly; as such, there is an increasing number of smooth defaults (not accompanied by large collateral shortfalls and hence not detected by our simplified definition of the PFE) that are not directly visible in the shown statistics.

In short, and with the obvious disclaimer that a forecasting model cannot be assessed based on a single observation, **we deem the model PFE levels broadly consistent with CS realised loss**. In the remainder of the paper, we are going to develop an analytical framework for WWR and leverage based on the model defined in Eqns.1-6.

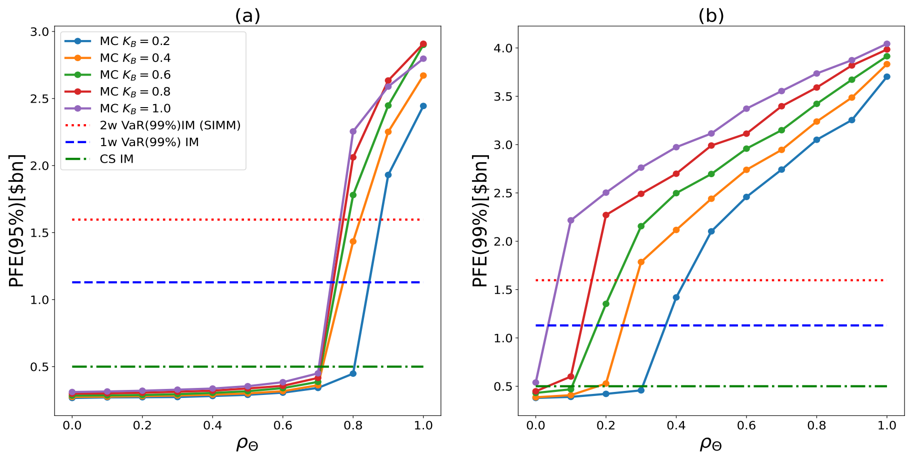

Figure 1: **(a) and (b)**: The model PFEs at the 95th **(a)** and 99th **(b)** quantiles computed using the Archegos calibration described in the main text.

5 The λ expansion methodology

The key piece of the puzzle is what we learned from the analysis in [1], i.e. that material collateral shortfalls are driven by large gap moves. Since the frequency of these moves λ_{J_S} tends to be a small number, we can perform a perturbative expansion in this parameter and compute $\mathrm{SF}_{\mathrm{VM}}$ by using the inputs from existing MC implementations (that generally do not include jump processes and defaults detection in simulation); i.e. without the need of implementing a MC framework based on Eqns.1-6, as we will see. For a first order expansion, the approximation should hold as long as PD $\gg \lambda_{J_S}^2$ and PD, $\lambda_{J_S} T \ll 1$.

$\mathrm{SF}_{\mathrm{VM}}$ is an expected shortfall conditional upon default, i.e. a quantity proportional to the ratio between the number of defaulting paths generating a shortfall and the total number of defaulting paths. As shown in Fig.2, in first order, there are three kinds of defaulting Θ paths to consider: J_1, $J_{1/W}$ and $J_{0/W}$; but only the former adds up to the collateral shortfall.

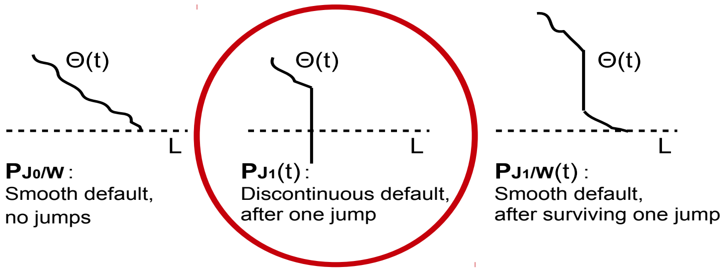

Figure 2: Diagrammatic illustration of the three kinds of $\Theta(t)$ paths that contribute in first order to $\mathrm{SF}_{\mathrm{VM}}$.

Making use of the fact that jumps are uniformly distributed over T, $\mathrm{SF}_{\mathrm{VM}}$ can be expressed in terms of J_1, $J_{1/W}$ and $J_{0/W}$ as:

$$\mathrm{SF}_{\mathrm{VM}}^{\lambda^1_{J_S}} = \frac{\int_0^T E_{J_1}(u)du}{P_{J_0/W} + \int_0^T P_{J_1}(u)du + \int_0^T P_{J_1/W}(u)du} \tag{11}$$

$$P_{J_0/W} = (1-\lambda_{J_S}T)\,\mathbb{E}\big[\mathbb{1}_{\{\mathrm{NAV}_{\min}(0,T)\leq 0\}}\big] \tag{12}$$

$$P_{J_1}(t) = \lambda_{J_S}\,\mathbb{E}\big[\mathbb{1}_{\{(\Theta(t^-)\leq\Theta_c)\wedge(\mathrm{NAV}_{\min}(0,t^-)>0)\}}\big] \tag{13}$$

$$E_{J_1}(t) = \lambda_{J_S}\,\mathbb{E}\big[-\Delta_t(g(t^+))\mathbb{1}_{\{(\Theta(t^-)\leq\Theta_c)\wedge(\mathrm{NAV}_{\min}(0,t^-)>0)\}}\big] \tag{14}$$

$$P_{J_1/W}(t) = \lambda_{J_S}\,\mathbb{E}\big[\mathbb{1}_{\{(\mathrm{NAV}_{\min}(t^+,T)\leq 0)\wedge(\Theta(t^-)>\Theta_c)\wedge(\mathrm{NAV}_{\min}(0,t^-)>0)\}}\big] \tag{15}$$

$$\Theta_c = \frac{L-\mathrm{MtM}(t^+)}{1+\rho_\Theta\mu_{J_S}} = \frac{L-V(t)-g(S(t^+),\sigma_S(t^+))}{1+\rho_\Theta\mu_{J_S}}, \tag{16}$$

where: i) the suffix $\lambda^1_{J_S}$ highlights the first order calculation; ii) the jump occurs at t; iii) $\mathbb{E}[]$ indicates the average across the continuous components of the simulated market paths for S, Θ, σ_S and any underlying of the non-WWR sub-portfolio V; iv) for a random process $X(t)$, $X_{\min}(t,T)$ indicates the minimum of $X(t)$ over the interval $[t,T]$, $X_{\min}(t,T) = \min_{u\in[t,T]}(X_u)$; and v) $-\Delta_t(g(t^+)) = g(S(t^-),\sigma_S(t^-)) - g(S(t^+),\sigma_S(t^+))$.

As a consequence of the perturbative expansion, we got rid of the jump stochastic dynamics; and the averages in Eqns.11-16 (and hence $\mathrm{SF}_{\mathrm{VM}}$) can be directly computed with standard CCR MC frameworks, without a full-blown implementation of Eqns.1-6. In practice, one can:

1. Value $V(t)$ and $g(S(t),\sigma_S(t))$ at different time steps using IMM or XVA scenarios from production models.

2. Simulate $\Theta(t)$ paths, using an additional GBM risk factor.

3. Perform post-jump valuations $g(S(t^+), \sigma_S(t^+))$ based on the same set of time steps and scenarios.

4. Compute $\mathrm{SF_{VM}}$ using Eqns.11-16.

Where the choice of the WWR calibration will affect the last three steps.

Eqns.11-16 constitute the first main result of the paper: we have shown that the $\mathbf{SF_{VM}}$ for arbitrary WWR pay-offs can be computed re-using IMM or XVA implementations and small additional trimmings. In the next section, we are going to solve Eqns.11-16 explicitly to obtain a closed-form expression for the EAD.

6 Calculation of the EAD

As for [1], we are primarily interested to the cases when the $\mathrm{SF_{VM}}$ contribution to the EAD is dominant compared to the smoother MPOR fluctuations in trades and collateral values accounted for by the Basel MPOR framework. Hence, we ignore the latter and we consider $\mathrm{EAD} \approx \mathrm{SF_{VM}}$. To proceed with the calculation, we make use of two key simplifications:

1. The maturity of the payoff function g is kept constant over the T=1y horizon, i.e. the CPTY rolls over continuously its initial position. This is consistent with the regulatory definition of the EAD (where roll-over risk is included via the effectivisation of the EE profile, see [12]); and, when comparing with MC simulations, we will use the roll-over assumption for both implementations.

2. The non-WWR portfolio distribution is modelled as a static random variable (RV), i.e. $V(t) = \tilde{V}$; where a static equivalent of a given non-WWR portfolio distribution $V(t)$ can be computed e.g. by gauging the average level of the $V(t)$ variance across the $[0, 1\mathrm{y}]$ interval.

Both shortcuts are required because of the same technical limitation, i.e. that a closed-form solution for the GBM hitting distribution (required to calculate analytically $P_{J_0/W}$ and $P_{J_1/W}$) is known only for constant (and few other) barriers.

In addition, and consistent with the above, we introduce some further simplifications: i) we ignore the conditioning $\mathrm{NAV_{min}}(0, t^-) > 0$ in Eqns.13, 14 and 15, since generally of order $(1 - \mathrm{PD}) \approx 1$; and ii) we ignore the drift terms μ_S and μ_Θ in Eqns.2 and 3 since of order λ_{J_S}.

Beside the points above, the main challenge is the handling of the *going concern* condition in Eqns.12 and 15 (i.e. before and after the jump); since NAV depends linearly on $\Theta(t)$ and $\tilde{V}$ and potentially non-linearly on $S(t)$ and $\sigma_S(t)$ (via the payoff function g). In this regard, we observe the following:

1. The RV $\tilde{V}$ can be considered as a random shift of L and will simply add an average across the $\tilde{V}$ scenarios.

2. If before and after the jump, the only stochastic dynamics is the one of the Brownian driver of the $\Theta(t)$ process, we can aim to express NAV as a function of $\Theta(t)$ only (plus the time-independent RV $\tilde{V}$); and the distribution of $\mathrm{NAV}_{\min}(0,t)$ becomes readily available since $\Theta(t)$ in between jumps is GBM. Accounting for the *going concern* condition then requires simply to find the roots of the equation $\mathrm{NAV}(\Theta)=0$.

3. Away from jumps, $S(t)$ and $\Theta(t)$ are correlated GBM processes. Hence, using an approximate Cholesky decomposition (that disregards fluctuations independent on Θ) and assuming a single jump occurring at $t=t^*$, we can express S as a function of Θ and make that NAV depends on Θ only (see previous point). $S(t)=S(\Theta(t))$ is given by the following equation (for more details, see App.C):

$$S(t) \approx S(0)\Theta(0)^{-\frac{\rho_W \sigma_S}{\sigma_\Theta}} \left(\frac{(1+\mu_{J_S})}{(1+\rho_\Theta \mu_{J_S})^{\frac{\rho_W \sigma_S}{\sigma_\Theta}}} \right)^{\theta(t-t^*)} \Theta(t)^{\frac{\rho_W \sigma_S}{\sigma_\Theta}}. \quad (17)$$

Based on these simplifications, we can rewrite $P_{J_0/W}$, $P_{J_1}(t)$, $E_{J_1}(t)$ and $P_{J_1/W}(t)$ as:

$$\begin{aligned}
P_{J_0/W} &= (1-\lambda_{J_S})\,\mathbb{E}_{\tilde{V}}\big[\mathbb{1}_{\{\mathcal{D}^-(\tilde{V})\}}\mathcal{M}(\Theta_L^-(\tilde{V}),\Theta_0,0,\sigma_\Theta,1\mathrm{y})\big] && (18)\\
P_{J_1}(t) &\approx \lambda_{J_S}\,\mathbb{E}\big[\mathbb{1}_{\{\mathcal{D}^+(\tilde{V})\}}\mathbb{1}_{\{\Theta(t^-)\le\Theta_c(\tilde{V})\}}\big] \\
&= \lambda_{J_S}\,\mathbb{E}_{\tilde{V}}\big[\mathbb{1}_{\{\mathcal{D}^+(\tilde{V})\}}\Phi^{LN}_{\Theta(t^-)}(\Theta_L^+(\tilde{V})/(1+\rho_\Theta\mu_{J_S}))\big] && (19)\\
E_{J_1}(t) &\approx \lambda_{J_S}\,\mathbb{E}\big[-\Delta_t(g(t^+))\mathbb{1}_{\{\Theta(t^-)\le\Theta_L^+(\tilde{V})/(1+\rho_\Theta\mu_{J_S})\}}\big] && (20)\\
P_{J_1/W}(t) &\approx \lambda_{J_S}\,\mathbb{E}\big[\mathbb{1}_{\{\mathcal{D}^+(\tilde{V})\}}\mathbb{1}_{\{\Theta_{\min}(t^+,1\mathrm{y})\le\Theta_L^+(\tilde{V})\}}\mathbb{1}_{\{\Theta(t^-)>\Theta_c\}}\big] && (21)\\
&= \lambda_{J_S}\,\mathbb{E}_{\tilde{V}}\big[\mathbb{1}_{\{\mathcal{D}^+(\tilde{V})\}}\int_{\Theta_L^+(\tilde{V})/(1+\rho_\Theta\mu_{J_S})}^{+\infty}\phi^{LN}_{\Theta(t^-)}(u)\,\times \\
&\qquad \mathcal{M}(\Theta_L^+(\tilde{V}),(1+\rho_\Theta\mu_{J_S})u,0,\sigma_\Theta,1\mathrm{y}-t)du)\big],
\end{aligned}$$

where: i) the jump occurs at t; ii) $\mathbb{E}[]$ is defined as for Eqns.11-16 (inclusive of the average over the non-WWR $\tilde{V}$ configurations) while $\mathbb{E}_{\tilde{V}}$ indicates only the average over $\tilde{V}$; iii) $\Theta_L^-(\tilde{V})$ and $\Theta_L^+(\tilde{V})$ are the (time-independent) largest positive roots of the NAV(Θ)=0 equation before and after the jump; iv) the indicator functions $\mathbb{1}_{\{\mathcal{D}^-(\tilde{V})\}}$ and $\mathbb{1}_{\{\mathcal{D}^+(\tilde{V})\}}$ define the domain of $\tilde{V}$ for which the default can occur (depending on the value of $\tilde{V}$, the VM collateral posted by A on a given scenario may exceed the liabilities L plus any shortfall driven by g); v) $\Phi^{LN}_{\Theta(t^-)}$ and $\phi^{LN}_{\Theta(t^-)}$ indicate the CDF and PDF of the $\Theta(t)$ lognormal distribution before the jump; and vi) $\mathcal{M}(x,\eta_t,\mu_\eta,\sigma_\eta,T-t)$

provides the likelihood for the minimum over the interval $[t,T]$ of a GBM process (μ_η, σ_η) with starting value η_t to be below x (see App.B and references therein).

Eqns.18-21 constitute the second main result of the paper, providing us with a closed-form expression for the EAD$\approx$ $\mathrm{SF_{VM}}$ for a generic WWR payoff. Recurring to a perturbative expansion in λ_{J_S} allowed us to reduce the calculation of $\mathrm{SF_{VM}}$, in principle involving a coupled system of SDEs with correlated random processes, to the computation of few expectations of lognormal distributions. As for Eq.17, we introduced an approximation for the S process, so as to express the *going concern* condition as a (generally) non-linear equation in Θ only.

Eqns.18-21 can turn useful e.g. in the context of stress testing, where the closed-form EAD can be used as a ready available desktop tool to gauge the impact of different levels of WWR on levered portfolios. [11].

In the next section, we will apply this result to the cases of two payoff examples: i) a long position on an Equity Total Return Swap (TRS) and ii) a short position on an Equity PUT option.

7 Results

7.1 MC benchmark

Fig.3(a) shows $\mathrm{SF_{VM}}$ for a long Equity TRS, computed using both MC and the analytical model for $\rho_W = 0.5$, i.e. for a level of diffusion correlation equidistant from the limits $\rho_W = 0$ and $\rho_W = 1$ when the approximation introduced in Eq. 17 should work worst and best respectively. As discussed more extensively App.D, the analytical model and MC simulations curves show the same qualitative behaviour (including all the crossovers between different K_B levels as a function of ρ_Θ); and the residual quantitative discrepancies are primarily driven by the above mentioned approximation and by the different margining frequencies (intraday vs. daily).

7.2 Non-linear payoffs

Fig.3(b) shows $\mathrm{SF_{VM}}$ for a short WWR OTM PUT with maturity T=1y, computed using both MC and the analytical model for the zero S diffusion case ($\beta_S = 0$). An OTM PUT is a yield enhancing position for the protection seller; and a prototypical non-linear WWR trade for the protection buyer, since it comes with the risk of having bought protection for scenarios that trigger the default of the protection seller.

Two points worth to remark: i) the SIMM style IM is practically 0 (the trade is OTM) and therefore provides no protection against the most likely default dynamics; and ii) for the same moneyness reason, the S diffusion has a minimal impact on the risk of the trade.

The latter point is shown more comprehensively in Fig.3(c)(inbox), where we compare $\mathrm{SF_{VM}}$ for the zero vs. non-zero diffusion cases as a function of the moneyness of the PUT. One can see that the two calibrations converge to the same $\mathrm{SF_{VM}}$ for deeply OTM positions, for which a gap move of the underlying is the only mechanism available to generate material exposure. Notice that the weak dependence on S diffusion for OTM products make things simpler; since, for $\beta_{\sigma_S} = 0$, the EAD can be obtained explicitly for any payoff and jump size (for more details, see App.E).

7.3 Concentration and WWR

Fig.3(d) shows $\mathrm{SF_{VM}}$ for a mixed portfolio WWR TRS + non-WWR V as a function of the concentration ω_c for $\rho_\Theta = 0.7$ and for different levels of K_B (for more details regarding the calculation, see App.F); where the average distribution $\tilde{V}$ has been extrapolated from the MPOR risk (fixed, given K_B and ω_c) as if V had a Brownian dynamics.

Few points worth to remark: i) concentration is a key risk factor for Archegos-like large shortfalls to materialise; as evident in Fig.3(d), IM is an adequate risk mitigant as long as *some* portfolio diversification is present; ii) the standard VM-collateralised EE (as computed according to the Basel MPOR framework) becomes of order $\mathrm{SF_{VM}}$ already for milder levels of concentration; iii) as a consequence of this, the total EAD (inclusive of both the shortfall at default and of the MPOR risk dynamics) is, in general, materially understated by the current standard.

Finally, Fig.3(e) shows the impact of the average moneyness of the non-WWR portfolio $\langle V \rangle = \mu_V L$, where $\mu_V L > 0$ increases $\mathrm{SF_{VM}}$. This can be understood in terms of the impact of V on the PD of CPTY B: a large and positive (negative) V depletes (augments) the PD materially by effectively reducing (increasing) the liability threshold L; and therefore depletes (augments) the relative weight of the smooth defaults in the average that determines $\mathrm{SF_{VM}}$.

8 Conclusions

We extended the model introduced in [1] for highly levered collateralised counterparties to address multiple cases, including linear and non-linear payoffs and mixed compositions of WWR and non-WWR trades. The model has been tested for the CS portfolio post-Archegos default and results were consistent with the observed closeout losses.

For the generalised model, we developed an analytical framework based on a perturbative expansion in the jump intensity λ_{J_S}; and this allowed us to express the expected collateral shortfall at default $\mathrm{SF_{VM}}$ as a set of expectations (Eqns.11-16), that can be computed using MC scenarios from existing IMM or XVA implementations.

Using the same analytical toolkit, we were able to obtain a closed-form solution for the EAD (Eqns.18-21) in the presence of WWR and leverage. This result can be applied e.g. in the context of non-simulation based risk frameworks, such as stress testing or SA-CCR.

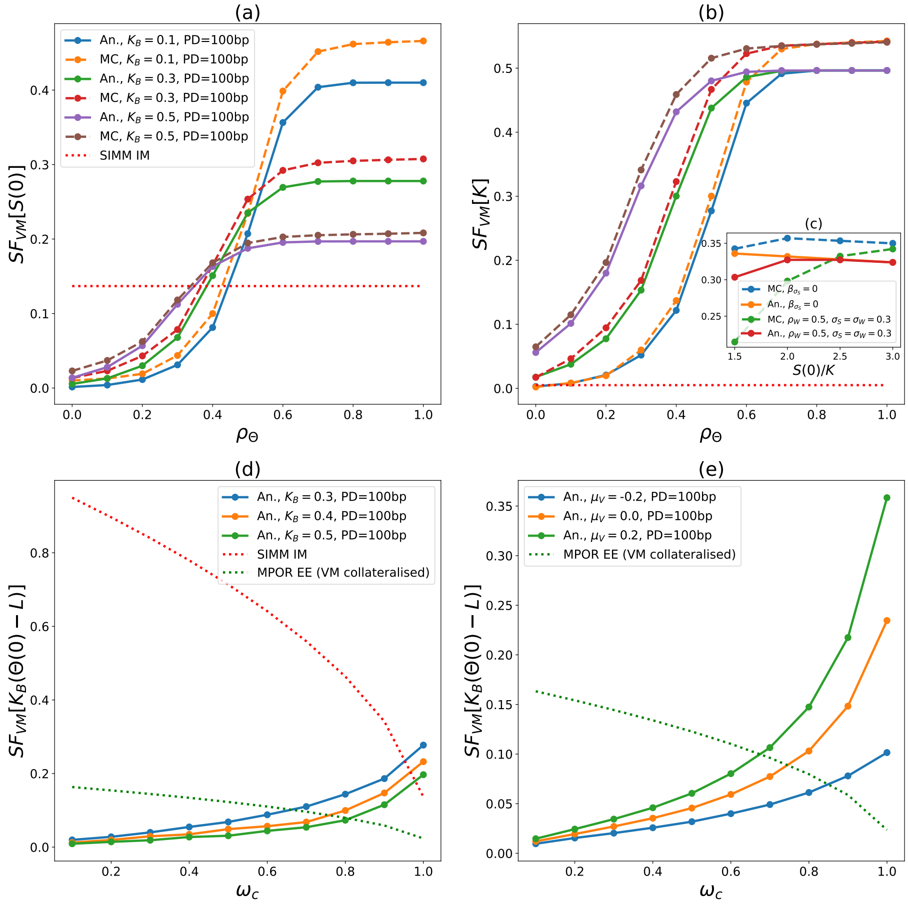

Figure 3: **(a) and (b)**: $\mathrm{SF}_{\mathrm{VM}}$ for a long TRS **(a)** and a short OTM PUT **(b)** computed with the analytical model and with MC is shown as a function of ρ_Θ for $\rho_W = 0.5$, $\sigma_S = \sigma_\Theta = 0.3$ and jump calibration JTD ($\lambda_{J_S} = 0.01$). **(c)**: $\mathrm{SF}_{\mathrm{VM}}$ for a sold PUT is shown as a function of the option moneyness for two calibrations,where in both cases: $\rho_\Theta = 0.3$, $K_B = 0.5$, jump calibration JTD ($\lambda_{J_S} = 0.01$) and PD=100bp. **(e) and (d)**: $\mathrm{SF}_{\mathrm{VM}}$ for a mixed portfolio WWR TRS + non-WWR V is shown as a function of ω_c for $\rho_W = 0.5$, $\sigma_S = \sigma_\Theta = 0.3$, $\rho_\Theta = 0.7$, jump calibration JTD ($\lambda_{J_S} = 0.01$) for different levels of: leverage K_{B} **(d)**; and moneyness of the non-WWR portfolio ($\tilde{V} = N(\mu_V \cdot L, \sigma_V)$, $K_B = 0.3$) **(e)**.

A Table of notations

Notation	Meaning
$S(t)$	WWR underlying
$g(S(t), \sigma_S(t))$	WWR payoff
$V(t)$	Non-WWR payoff
$\Theta(t)$	Balance sheet of the defaulting cpty B
$\mathrm{NAV}(t)$	Net asset value of the defaulting cpty B
L	$t = 0$ liabilities of the defaulting cpty B
K_B	Riskiness of the portfolio in units of the available capital
ω_c	Concentration of the WWR portfolio
g_{JTD}	WWR payoff valued for $S = 0$
μ_S, σ_S	Drift of and volatility of the diffusion component of $S(t)$
μ_Θ, σ_Θ	Drift of and volatility of the diffusion component of $\Theta(t)$
λ_{J_S}, μ_{J_S}	Intensity and jump size of the compound Poisson process J_S
β_S	(0,1) binary switch for the diffusion component of $S(t)$
ρ_W	Instantaneous correlation between the Brownian drivers of $S(t)$ and $\Theta(t)$
ρ_Θ	WWR coupling that projects the $S(t)$ jump component on $\Theta(t)$
ρ_{σ_S}	WWR coupling that projects the $S(t)$ jump component on $\sigma_S(t)$
$\mathrm{SF}_{\mathrm{VM}}$	Expected collateral shortfall at default
$\Delta_{1\mathrm{d}}(\mathrm{MtM})$	The daily accrued change in MtM based on EOD market quotes

B Distribution of the minimum of GBM over an interval $[0, T]$

$$\mathcal{M}(x, x_0, \mu_x, \sigma_x, T) = \Phi^N(X^-) + \left(\frac{x_0}{x}\right)^{1-2\mu_x/\sigma_x^2} \Phi^N(X^+) \quad (22)$$

$$X^\pm = \frac{\ln(x/x_0) \pm (\mu_x - \sigma_x^2/2)T}{\sigma_x\sqrt{T}}, \quad (23)$$

where μ_x and σ_x indicate the drift and volatility of the given GBM process. For the details of the derivation of the formula above, we refer e.g. to [13].

C Approximate Cholesky decomposition

In the following we use a Cholesky decomposition for S where: i) W_1 is the Brownian driver of Θ and ii) W_2 is a second Brownian driver independent of W_1. Assuming at most one jump at $t = t^*$, we can express S as:

$$\begin{aligned} S(t) &= S(0)\Theta(0)^{-\frac{\rho_W\sigma_S}{\sigma_\Theta}}(1+\mu_{J_S})^{\theta(t-t^*)} e^{-(1-\rho_W^2)\sigma_S^2 t/2+\sqrt{1-\rho_W^2}W_2(t)} \times \\ &\quad (1+\rho_\Theta\mu_{J_S})^{-\frac{\rho_W\sigma_S}{\sigma_\Theta}\theta(t-t^*)} e^{-\rho_W^2\sigma_S^2 t/2} e^{\rho_W\sigma_S\sigma_\Theta t/2}\Theta(t)^{\frac{\rho_W\sigma_S}{\sigma_\Theta}}. \end{aligned} \quad (24)$$

To derive Eq.17, the main approximation used is to average out the fluctuations independent on Θ, i.e.:

$$\begin{aligned} S(t) &\approx S(0)\Theta(0)^{-\frac{\rho_W\sigma_S}{\sigma_\Theta}}(1+\mu_{J_S})^{\theta(t-t^*)}\,\mathbb{E}_{W_2(t)}\left[e^{-(1-\rho_W^2)\sigma_S^2 t/2+\sqrt{1-\rho_W^2}W_2(t)}\right]\times \\ &\quad (1+\rho_\Theta\mu_{J_S})^{-\frac{\rho_W\sigma_S}{\sigma_\Theta}\theta(t-t^*)}e^{-\rho_W^2\sigma_S^2 t/2}e^{\rho_W\sigma_S\sigma_\Theta t/2}\Theta(t)^{\frac{\rho_W\sigma_S}{\sigma_\Theta}} \qquad (25) \\ &= S(0)\Theta(0)^{-\frac{\rho_W\sigma_S}{\sigma_\Theta}}\left(\frac{(1+\mu_{J_S})}{(1+\rho_\Theta\mu_{J_S})^{\frac{\rho_W\sigma_S}{\sigma_\Theta}}}\right)^{\theta(t-t^*)}\Theta(t)^{\frac{\rho_W\sigma_S}{\sigma_\Theta}}\times \\ &\quad e^{-\rho_W^2\sigma_S^2 t/2+\rho_W\sigma_S\sigma_\Theta t/2}, \qquad (26) \end{aligned}$$

and, in addition, to ignore the time-dependent factor $e^{-\rho_W^2\sigma_S^2 t/2+\rho_W\sigma_S\sigma_\Theta t/2}\approx 1$; so as to obtain the desired expression:

$$S(t)\approx S(0)\Theta(0)^{-\frac{\rho_W\sigma_S}{\sigma_\Theta}}\left(\frac{(1+\mu_{J_S})}{(1+\rho_\Theta\mu_{J_S})^{\frac{\rho_W\sigma_S}{\sigma_\Theta}}}\right)^{\theta(t-t^*)}\Theta(t)^{\frac{\rho_W\sigma_S}{\sigma_\Theta}}. \qquad (27)$$

Regarding the approximations introduced: i) the contribution from the W_2 fluctuations becomes 0 in the limit $|\rho_W|\to 1$ (Eq.17 is exact for $\rho_W=1$ and $\sigma_S=\sigma_\Theta$) while, for $|\rho_W|\to 0$, the approximation is doomed to fail since W_2 is the sole driver of the variance of S; ii) for $t\leq$1y, as long as $\sigma_S\approx\sigma_\Theta$, the term $(-\rho_W^2\sigma_S^2 t/2+\rho_W\sigma_S\sigma_\Theta t/2)$ is generally small for any $\rho_W>0$.

D MC benchmark

The analytical expansion introduced in Secs.5 and 6 make use of three main approximations compared to the model described by Eqns.1-6, namely:

A1: The continuous time margining (vs. daily for Eqns.1-6), as discussed in Sec.3.

A2: The first order λ expansion, as discussed in Sec.5.

A3: The approximate Cholesky decomposition, as discussed in App.C.

For the parameters domain considered, A2 is always fulfilled; hence, the residual discrepancy between the MC simulation of Eqns.1-6 and the analytical model is driven by A1 and A3.

Fig.4(a) (long TRS) and Fig.3(b) in the main text (short OTM PUT) show a comparison between the $\mathrm{SF}_{\mathrm{VM}}$ calculated: i) simulating Eqns.1-6 with MC and ii) using the analytical expansion of Eqns.18-21. For such cases, S diffusion is absent ($\beta_{\sigma_S}=0$) and A3 does not apply. Therefore, the discrepancy between the MC and analytical results is directional and solely driven by A1. Referring to the diagrams shown in Fig.1 of the main text, the margining frequency primarily impacts $P_{J_0/W}$ and $P_{J_1/W}(t)$, where the

CDF $\mathcal{M}(x, \eta_t, \mu_\eta, \sigma_\eta, T-t)$ provides for the same minimum $\Theta = x$ a larger likelihood compared to the discrete time equivalent (based on measuring Θ only on a discrete daily grid). Hence, since both $P_{J_0/W}$ and $P_{J_1/W}(t)$ appear in the denominator of Eq.11, the continuous-time analytical model produces a lower $\mathrm{SF}_{\mathrm{VM}}$ compared to the daily margined MC simulation of Eqns.1-6.

Fig.4(b) shows the $\mathrm{SF}_{\mathrm{VM}}$ of a long TRS in the presence of S diffusion ($\beta_{\sigma_S} = 1$ and $\sigma_S = \sigma_\Theta = 0.3$) and for different levels of the diffusion correlation ρ_W. In this case, the discrepancy between the MC and analytical results is driven by both A1 and A3. Where: i) the two approximations contribute with opposite signs; and ii) A3 becomes dominant for $\rho_W \to 0$.

Finally notice that, for the purpose of modelling WWR portfolios, we can expect in general that $|\rho_W| \gg 0$. As shown in Figs.3(a) and 3(c) and discussed in the main text, for the examples considered and $\rho_W = 0.5$, the combined impact of A1 and A3 is fairly limited.

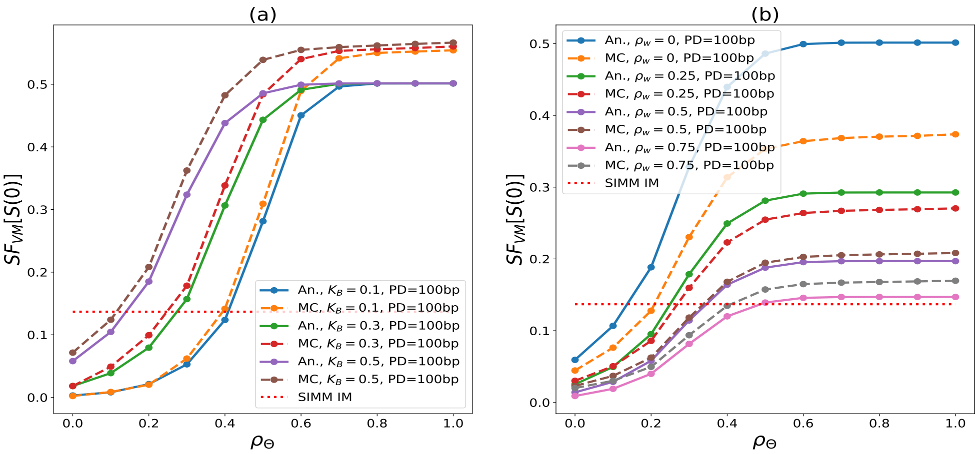

Figure 4: **(a)**: $\mathrm{SF}_{\mathrm{VM}}$ for a long TRS is shown as a function of ρ_Θ for $\beta_S = 0$ (i.e. no S diffusion), jump calibration JTD, PD=100bp and different levels of K_{B}. **(b)**: $\mathrm{SF}_{\mathrm{VM}}$ for a long TRS is shown as a function of ρ_Θ for $\sigma_S = \sigma_\Theta$, jump calibration JTD, PD=100bp and different levels of ρ_W.

E EAD in the absence of S diffusion ($\beta_{\sigma_S} = 0$)

In the absence of S diffusion ($\beta_S = 0$), $g(t^-)$ and $g(t^+)$ do not depend on Θ and the equation NAV(Θ)=0 is linear. Therefore, its roots before and after the jump $\Theta_L^-(\tilde{V})$ and $\Theta_L^+(\tilde{V})$ can be determined straightforwardly as:

$$\begin{aligned}\Theta_L^-(\tilde{V}) &= L - \tilde{V} - g(t^-) \\ &= L - \tilde{V} - g(S(0), \sigma_S(0)) \quad (28) \\ \Theta_L^+(\tilde{V}) &= L - \tilde{V} - g(t^+) \\ &= L - \tilde{V} - g(S(0)(1+\mu_{J_S}), \sigma_S(0)(1+\rho_{\sigma_S})); \quad (29)\end{aligned}$$

where (see Eq.4) we assumed that $\sigma_S(0) = \bar{\sigma}_S$ and, as discussed in the main text, we ignored the mean reversion dynamics post jump. Having $\Theta_L^-(\tilde{V})$ and $\Theta_L^+(\tilde{V})$, Eqns.18-21 can be directly computed for a generic payoff g and jump size μ_{J_S} and can be expressed as:

$$\begin{aligned}\mathrm{SF}_{\mathrm{VM}} &= \frac{-\Delta_t(g(t^+)) \int_0^{1\mathrm{y}} P_{J_1}(t)dt}{P_{J_0/W} + \int_0^{1\mathrm{y}} P_{J_1}(t)dt + \int_0^{1\mathrm{y}} P_{J_1/W}(t)dt} \quad (30) \\ P_{J_0/W} &= (1-\lambda_S)\,\mathbb{E}_{\tilde{V}}\left[\mathbb{1}_{\{L>\tilde{V}+g(t^-)\}}\mathcal{M}(L-\tilde{V}-g(t^-), \Theta_0, 0, \sigma_\Theta, 1\mathrm{y})\right] \quad (31) \\ P_{J_1}(t) &\approx \lambda_S\,\mathbb{E}_{\tilde{V}}\left[\mathbb{1}_{\{L>\tilde{V}+g(t^+)\}}\Phi^{LN}_{\Theta(t^-)}(\Theta_c(\tilde{V}))\right] \quad (32) \\ P_{J_1/W}(t) &= \lambda_S\,\mathbb{E}_{\tilde{V}}\left[\mathbb{1}_{\{L>\tilde{V}+g(t^+)\}}\int_{\hat{\Theta}_c}^{+\infty}\phi^{LN}_{\Theta(t^-)}(U-\tilde{V}/(1+\rho_\Theta\mu_{J_S}))\times\right. \\ &\left.\mathcal{M}(L-\tilde{V}-g(S(t^+),\sigma_S(t^+)), (1+\rho_\Theta\mu_{J_S})U-\tilde{V}, 0, \sigma_\Theta, 1\mathrm{y}-t)dU)\right] \\ \Theta_c(\tilde{V}) &= \frac{L-\tilde{V}-g(S(t^+),\sigma_S(t^+))}{1+\rho_\Theta\mu_{J_S}} = \hat{\Theta}_c - \frac{\tilde{V}}{1+\rho_\Theta\mu_{J_S}}. \quad (33)\end{aligned}$$

F EAD in the presence of S diffusion: analytical solution for JTD and $\frac{\sigma_S\rho_W}{\sigma_\Theta} = 0.5$

Addressing the general case of non-zero S diffusion requires the usage of a numerical root finder to solve the NAV(Θ)=0 equation. In this section, we consider a special case for which the equation admits a simple analytical solution, i.e.:

(i) JTD jump calibration.

(ii) $\frac{\sigma_S\rho_W}{\sigma_\Theta} = 0.5$.

(iii) Linearised payoff $g(S(t), \sigma_S(t)) \approx g(S(0)) + \Delta_g(S(t) - S(0))$ (exact for linear trades such as TRS).

As for the previous section, the main step is to determine $\Theta_L^-(\tilde{V})$ and $\Theta_L^+(\tilde{V})$. Before the jump, we make use of the linearised payoff to write the NAV($\Theta(t)$)=0 equation as:

$$\Theta(t) + g(S(0)) + \Delta_g S(0)(\Theta(0)^{-\frac{\rho_W\sigma_S}{\sigma_W}}\Theta(t)^{\frac{\rho_W\sigma_S}{\sigma_\Theta}} - 1) + \tilde{V} - L = 0; \quad (34)$$

where we are only interested in the root of Eq. 34 that can assume a positive value, i.e.:

$$\Theta_L^-(\tilde{V}) = \frac{1}{4}\left(-\frac{\Delta_g S(0)}{\Theta^{\frac{1}{2}}(0)} + \sqrt{\left(\frac{\Delta_g S(0)}{\Theta^{\frac{1}{2}}(0)}\right)^2 - 4(g(S(0)) + \tilde{V} - L - \Delta_g S(0))} \right)^2 . \tag{35}$$

The condition $\Theta_L^-(\tilde{V}) > 0$ can be read out from Eq.35 above as:

$$(g(S(0)) + \tilde{V} - L - \Delta_g S(0)) < 0; \tag{36}$$

and Eq.36 defines the domain $\mathcal{D}^-(\tilde{V})$ of $\tilde{V}$ values in Eqns.18-21 for which a default can occur. After the jump, the root $\Theta_L^+(\tilde{V})$ is trivial, since unaffected by the S dynamics (as a result of the JTD calibration) and given by:

$$\Theta_L^+(\tilde{V}) = L - \tilde{V} - g_{\text{JTD}}, \tag{37}$$

where g_{JTD} indicates the g payoff upon S default.

Having $\Theta_L^-(\tilde{V})$ and $\Theta_L^+(\tilde{V})$, the only residual technicality is the computation of $E_{J_1}(t)$, that is now affected by the S dynamics before the jump. Performing the average over the S configurations, $E_{J_1}(t)$ can be written as:

$$\begin{aligned} E_{J_1}(t) &= \lambda_S \, \mathbb{E}\big[-\Delta_t(g(t^+))\mathbb{1}_{\{L>\tilde{V}+g(t^+)\}}\mathbb{1}_{\{\Theta(t^-)\leq\Theta_c(\tilde{V})\}}\big] && (38)\\ &= X_1 + X_2 \\ X_1 &= (-g_{\text{JTD}} + g(0) - \Delta_g S(0)) * \\ &\quad\ \lambda_S \, \mathbb{E}_{\tilde{V}}\big[\mathbb{1}_{\{L>\tilde{V}+g(t^+)\}}\Phi^{LN}_{\Theta(t^-)}(\Theta_c(\tilde{V}))\big] && (39)\\ X_2 &= \Delta_g S_0 * \lambda_S \, \mathbb{E}_{\tilde{V}}\big[\mathbb{1}_{\{L>\tilde{V}+g(t^+)\}}\Phi^{LN}_{\Theta(t^-)}(\Theta_c(\tilde{V})/e^{\rho_W\sigma_S\sigma_\Theta t})\big]. && (40) \end{aligned}$$

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